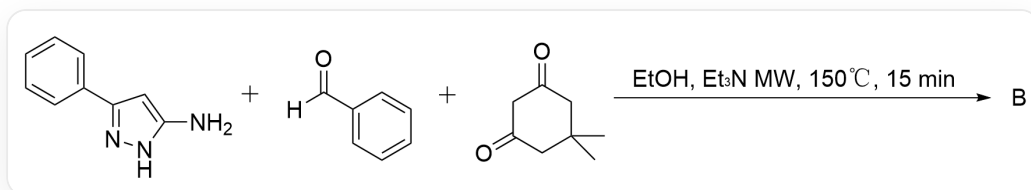


Question

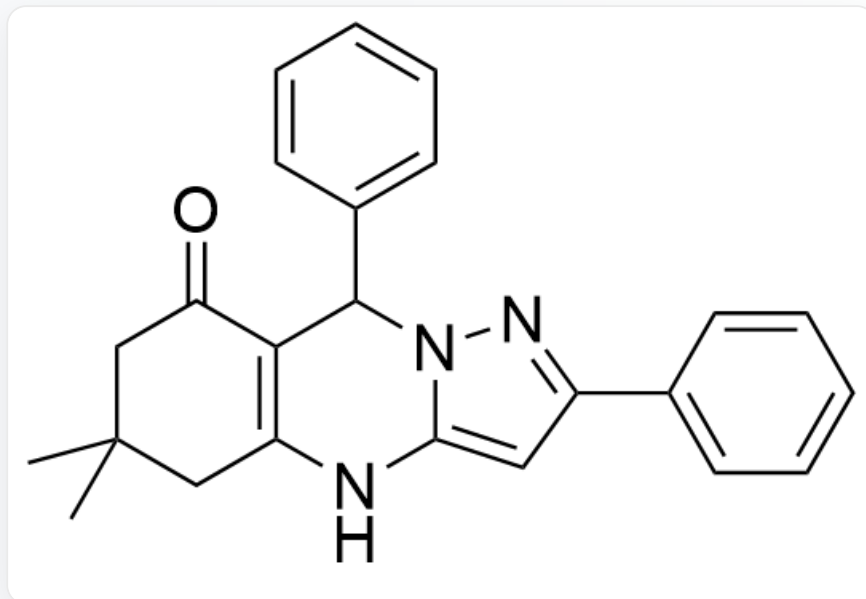
Microwave catalysis is based on the unique thermal effects generated by the principle of dielectric heating, which can lead to a significant increase in reaction rates.



NC1=CC(C2=CC=CC=C2)=NN1.O=C(C1=CC=CC=C1)[H].O=C1CC(C)(C)CC(C1)=O>EtOH, Et3N MW, 150°C, 15 min>B

Please attempt to predict the product B of the above reaction and indicate how many stereocenters the product possesses.

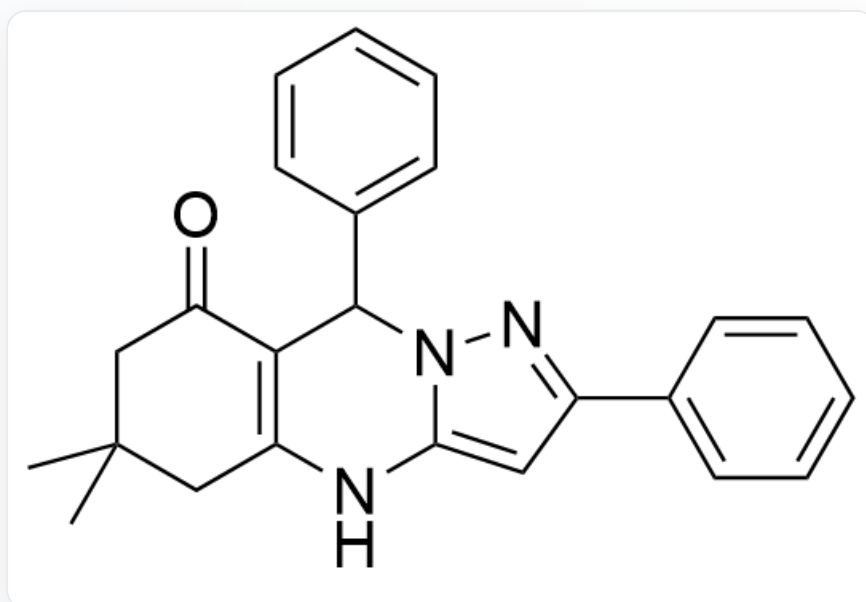
A.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

There are 2 chiral centers

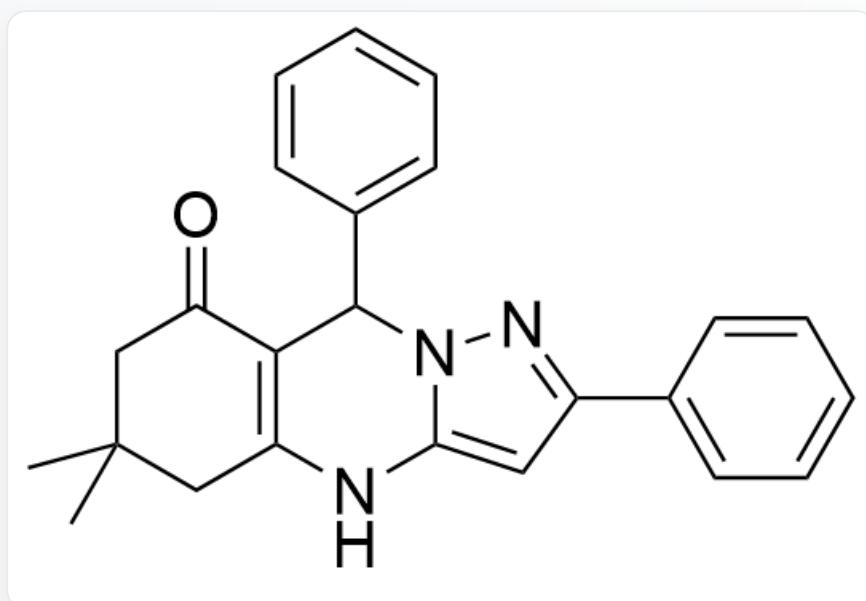
B.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

There is 1 chiral center

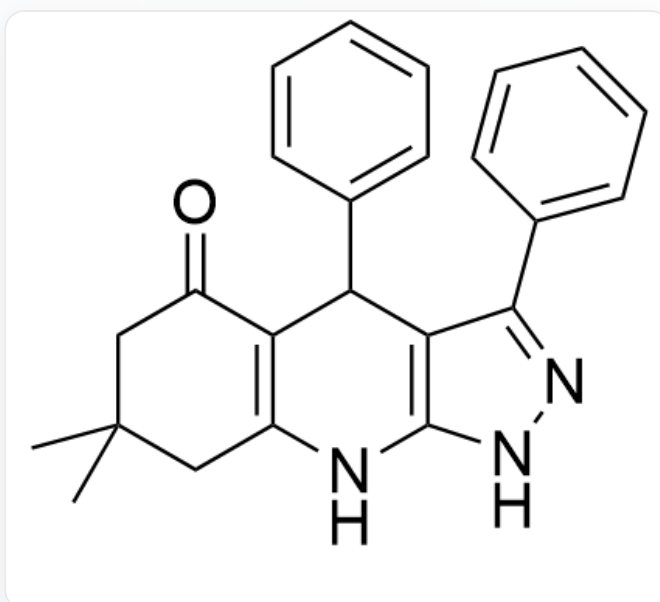
C.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

There are 0 chiral centers

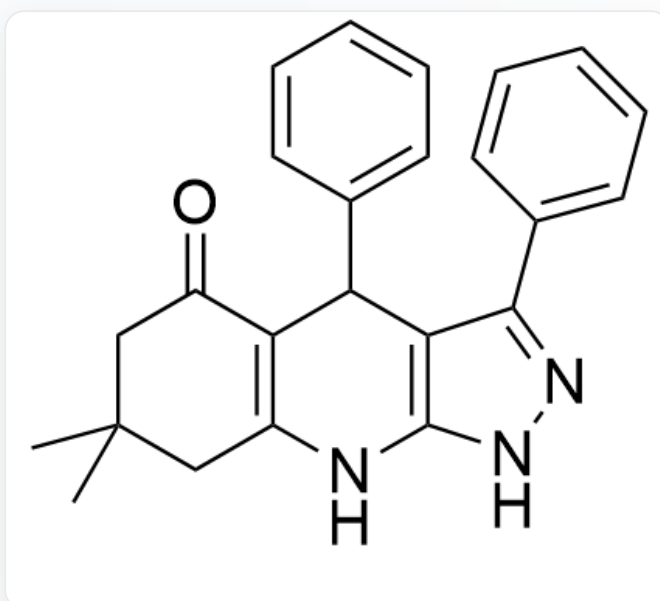
D.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

There are 2 chiral centers

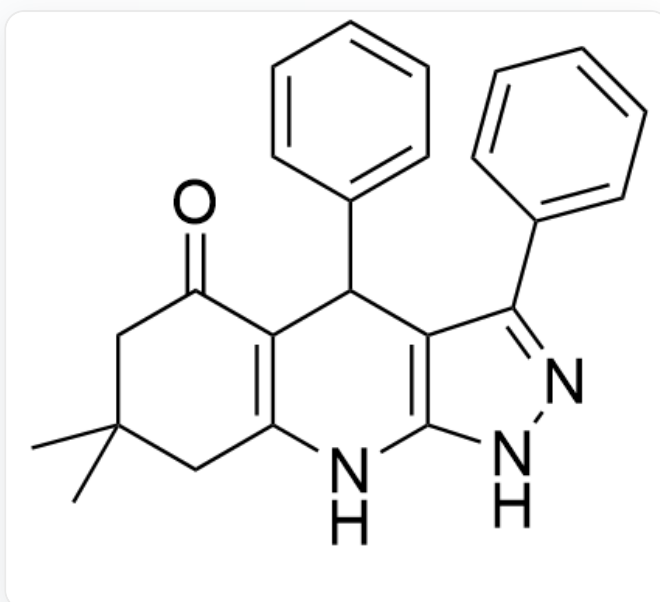
E.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

There is 1 chiral center

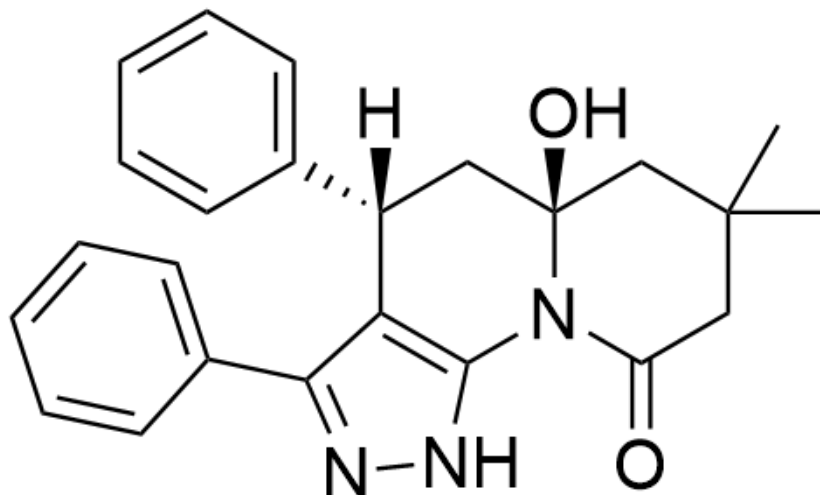
F.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

There are 0 chiral centers

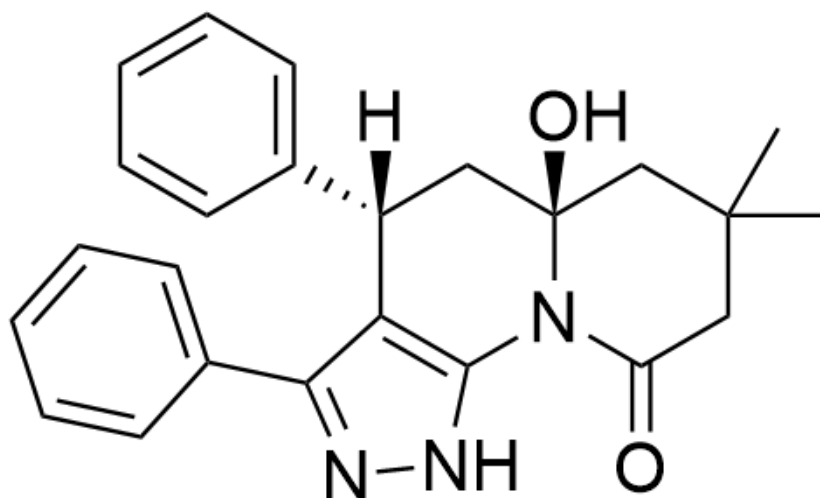
G.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 2 chiral centers

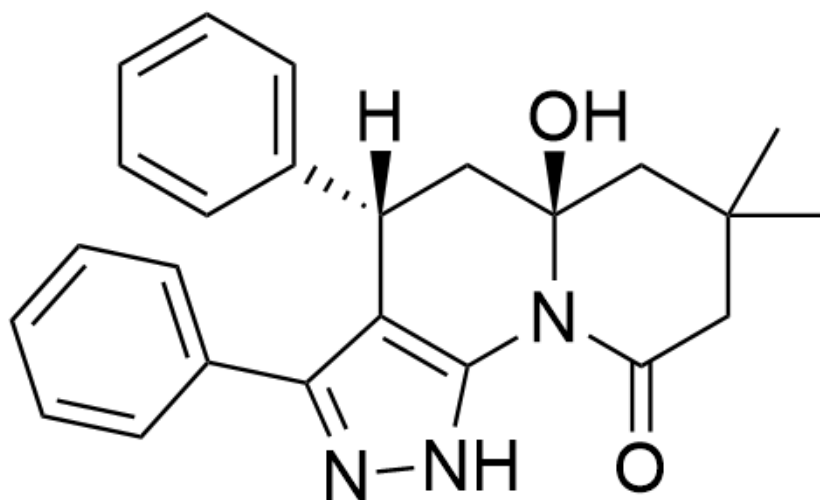
H.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There is 1 chiral center

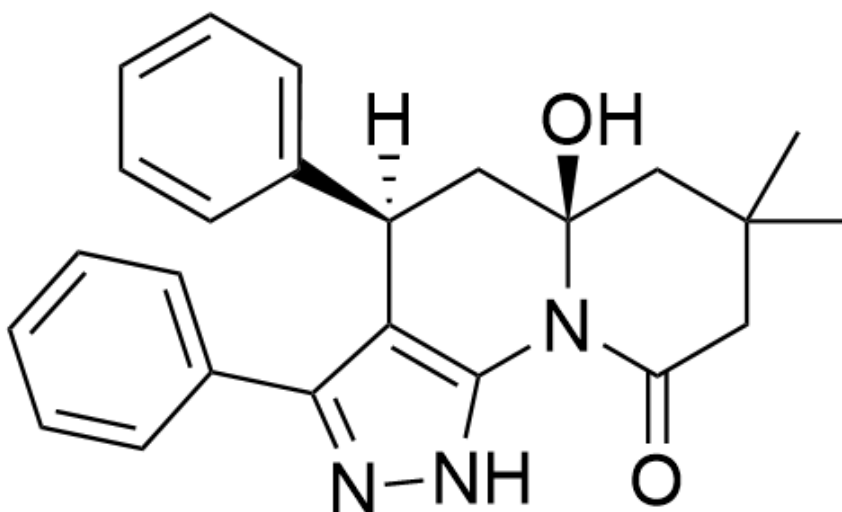
I.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 0 chiral centers

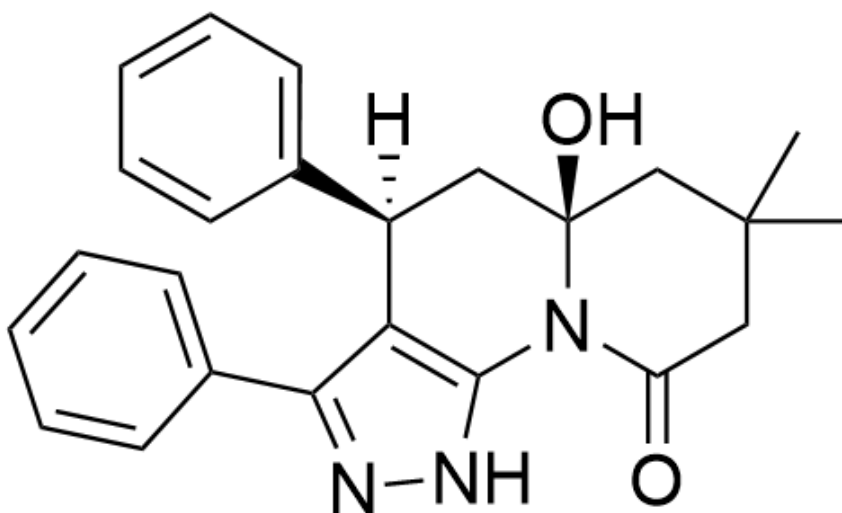
J.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 2 chiral centers

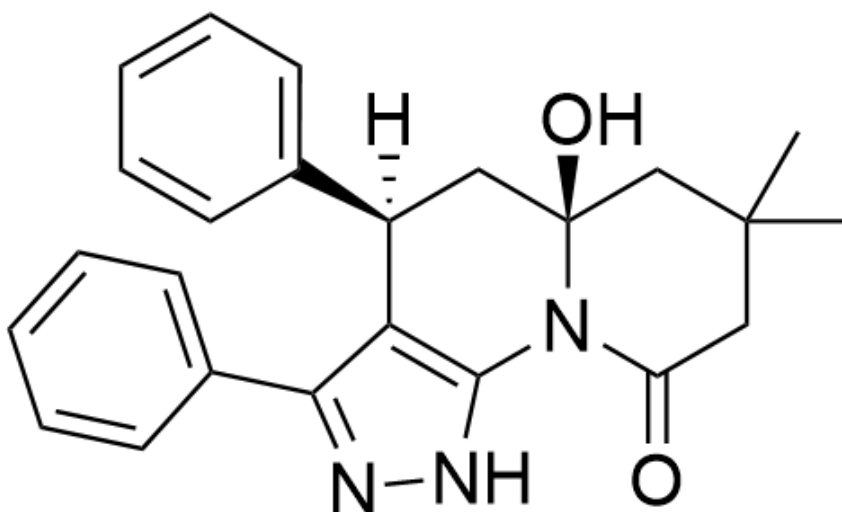
K.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There is 1 chiral center

L.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

There are 0 chiral centers

Answer

Correct Answer: **E**

Detailed Explanation

Under the conditions of adding triethylamine and a reaction temperature of 150°C, the reaction yields the thermodynamic product.

CHECKPOINT

1 PTS

The reaction yields the thermodynamic product

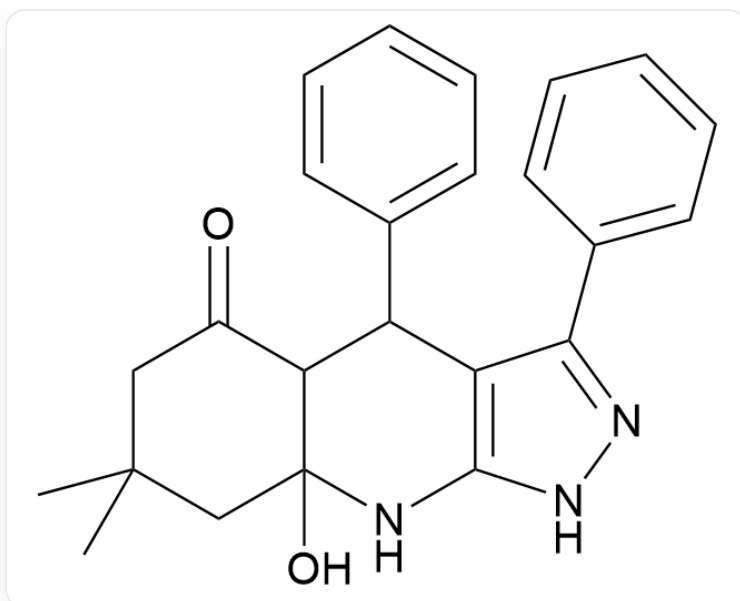
The formation of a carbon-carbon single bond during ring closure is thermodynamically more stable than the formation of a carbon-nitrogen single bond.

CHECKPOINT

1 PTS

The formation of a carbon-carbon single bond during ring closure is thermodynamically more stable than the formation of a carbon-nitrogen single bond

First, the condensation reaction intermediate is obtained

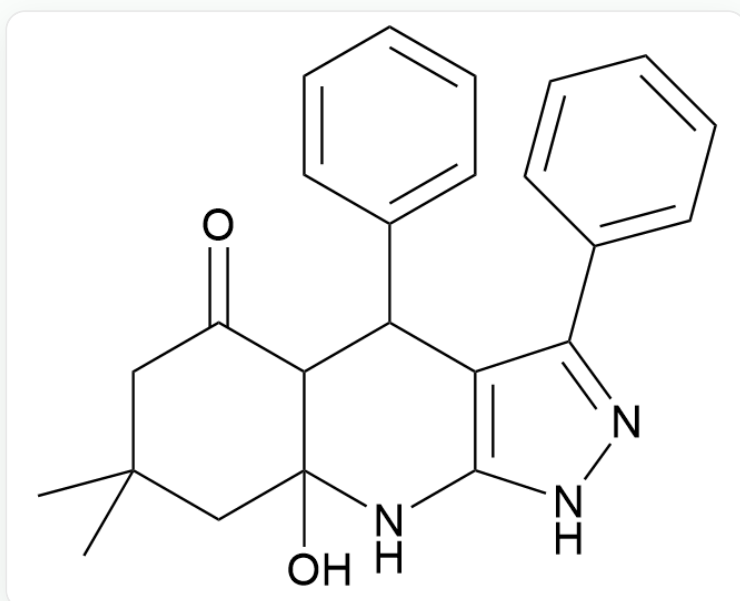


O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

CHECKPOINT

1 PTS

First, the condensation reaction intermediate is obtained



O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

Finally, a molecule of water is lost to yield the final product E.

Clearly, this product has only one stereocenter.